Hypoglycemia Reshapes Stratified Neural Dynamics Without a Confirmed Reachability Collapse

A paired reanalysis of 34 type 1 diabetes EEG recordings

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Abstract Acute hypoglycemia can increase slow EEG activity while impairing cognition, making rhythmic structure alone an ambiguous index of functional neural organization. We reanalyzed public 19-channel EEG from 34 people with type 1 diabetes studied during euglycemic and hypoglycemic clamp excerpts. A fixed NeuralManifoldDynamics contract summarized each five-minute condition with 3D and stratified 9D state coordinates, local Jacobian metrics, and short-horizon reachability. In paired hypoglycemia minus euglycemia contrasts, 3D mobility decreased (mean delta = −0.743, FDR q = 0.00090). In the 9D chart, mobility components m_a and m_e and entropy-related e_e decreased, whereas d_n and e_m increased (all q <= 0.0048). Raw EEG sanity checks replicated higher theta and delta power and lower exported entropy measures. Reachability effective dimension and directional entropy were nominally higher under hypoglycemia, but these effects did not survive the broader engine-level reachability correction; reachability log-volume was not significant. A 10,000-draw paired label-swap test and exclusion of three adjusted-geometry hypoglycemia files preserved the directions of all prespecified 9D and reachability effects. These results establish a bounded condition-associated stratified state-space signature, not a glucose dose-response, a recovery trajectory, or a causal metabolic gate.

Author Summary

Low blood glucose changes the EEG, but a larger slow rhythm does not automatically mean better neural coordination. In a paired diabetes EEG dataset, hypoglycemia altered several components of a multi-part neural-dynamics chart while broad reachability-collapse claims were not supported. The result shows why metabolic perturbation must be separated from any direct inference about neural capacity, cognition, or consciousness.

1. Introduction

Acute hypoglycemia is a clinically important metabolic perturbation that can alter cognition and surface electrophysiology. Its cognitive consequences are heterogeneous across clinical histories and exposure regimes, but severe hypoglycemia is associated with adverse cognitive outcomes in important subgroups (Tupola & others, 2020). Quantitative EEG studies show that insulin-induced hypoglycemia can increase low-frequency activity (Reno & others, 2015). This creates an interpretation problem for dynamical accounts that treat rhythmic regularity as intrinsically beneficial.

Rubega and colleagues released 19-channel EEG from 34 people with type 1 diabetes recorded in paired euglycemic and hypoglycemic clamp excerpts (Rubega et al., 2020). Their analysis established condition-sensitive nonlinear complexity features. We use the same bounded design to ask whether a fixed multicomponent neural-dynamics measurement contract differentiates the two metabolic conditions, and whether it supports a claim of reduced short-horizon reachability.

Core question

Does experimentally induced hypoglycemia produce a reproducible paired shift in a fixed 3D/9D neural-dynamics chart, and does that shift justify a claim of reduced short-horizon reachability?

2. Theory

2.1. Measurement-chart framing

Let X_t denote an unobserved neural process and Y_t the recorded EEG. The analysis does not identify X_t; it constructs a versioned chart x_t from Y_t:

$$Y_t = g(X_t) + \varepsilon_t, \quad x_t = C(Y_t).$$

The chart has a coarse representation (m, d, e) and a stratified representation $(m_a, m_e, m_o, d_n, d_l, d_s, e_e, e_s, e_m)$. These coordinates are operational feature aggregates, not unique latent variables. Local flow is summarized by an estimated Jacobian J_t, while reachability summarizes the covariance of nearby model-propagated futures:

$$\Sigma_t^{h+1} = A_t \Sigma_t^h A_t^T + Q_t.$$

Realized coordinates, local flow, and short-horizon capacity are therefore distinct layers. None is interpreted as a direct measure of consciousness.

2.2. Scope boundary: a metabolic perturbation, not embodied anchoring

This analysis does not test the Embodied Anchoring Principle. The pipeline exports no cardiac, respiratory, heartbeat-evoked, autonomic, ownership, or interoceptive-awareness variable, and its paired contrast is solely hypoglycemia minus euglycemia. Metabolic state is therefore treated here as an experimentally controlled perturbation condition, not as an embodied anchor, a rhythmic phase signal, or a direct variable of conscious level.

A future study with continuous glucose, rate of change, autonomic signals, behavior, and recovery could test whether metabolic availability constrains a neural-dynamics model. The present excerpts cannot estimate a metabolic adequacy factor, identify a causal gate on local geometry, or distinguish metabolic state from its interoceptive representation.

3. Methods

3.1. Dataset

We reanalyzed the public Rubega et al. type 1 diabetes clamp EEG dataset (Rubega et al., 2020) (Zenodo 3465213). It contains 34 participants with 19-channel scalp EEG sampled at 200 Hz. Each participant contributed two separately excerpted five-minute recordings: euglycemia and hypoglycemia. The release contains neither a common continuous timeline between excerpts nor participant-level glucose, awareness, demographic, recovery, or behavioral measures.

3.2. NeuralManifoldDynamics pipeline

The source H5 bundle was generated on 23 July 2026 under the identifier `neuralmanifolddynamics_hypoglycemia_t1d_20260723_180515`. EEG was filtered from 1 to 45 Hz, notch filtered at 50 Hz, and average rereferenced according to the embedded ingest configuration. The pipeline used 8-second windows at 4-second steps, yielding 74 windows per condition and participant. It exported 3D and 9D coordinates, 3D and 9D Jacobians, raw EEG features, and subject- and cohort-anchored coordinate layers.

The primary analysis uses the cohort-anchored layer. Its frozen feature anchor was fit from pooled euglycemia and hypoglycemia rows, so it supplies a shared comparison frame but is not an euglycemia-only metabolic baseline.

3.3. Endpoints and statistical design

Primary chart endpoints were 3D m, d, and e medians; nine stratified coordinate medians; 3D Jacobian Frobenius norm, rotation norm, and trace; and reachability log-volume, effective dimension, condition number, directional entropy, and rotation. Raw delta, theta, alpha, sample/permutation entropy, and Hjorth complexity served as secondary feature-level sanity checks.

The unit of inference was the participant-condition summary. We tested HYPO minus EU with the pipeline's paired permutation procedure (5,000 permutations) and bootstrap confidence intervals. Benjamini–Hochberg FDR controlled multiplicity within contrast and analysis type (Benjamini & Hochberg, 1995). The engine reports `right - left`; the final configuration therefore explicitly defined euglycemia as left and hypoglycemia as right.

3.4. Robustness protocol

All 34 pairs were complete for 3D/9D coordinate analyses. Thirty-three pairs were testable on the reachability surface because one 9D Jacobian export was unavailable. We added a 10,000-draw within-participant label-swap null over 11 prespecified 9D/reachability endpoints. A second sensitivity analysis excluded three hypoglycemia files whose embedded geometry contract was marked `adjusted` (`sub-004`, `sub-017`, and `sub-021`). These checks assess association stability, not unmeasured physiological confounding.

4. Results

4.1. R1: Data quality and coverage

The input bundle contained 68 readable H5 files (34 participant pairs). Every file provided finite 3D and 9D coordinates and 74 time windows. The 3D Jacobian was available for all pairs. One hypoglycemia 9D Jacobian was absent, leaving 33 reachability pairs. No regional, continuous-time, or embodied-anchor exports were present.

Surface	EU	HYPO	Paired N
3D and 9D coordinates	34	34	34
3D Jacobian	34	34	34
Reachability	33	33	33

Table 1: Analyzable participant-condition summaries after contract-based QC.

4.2. R2: Coarse and stratified coordinates

Hypoglycemia was associated with lower coarse mobility ($\Delta m = -0.743$, $q = 0.00090$). Coarse d did not differ after correction and the coarse e decrease did not survive FDR ($q = 0.0747$). The stratified chart resolved a mixed pattern: m_a, m_e, and e_e were lower under hypoglycemia, while d_n and e_m were higher. The condition effect is therefore not a uniform shift along a single coarse coordinate.

4.3. R3: Local flow and reachability

None of the three prespecified 3D Jacobian endpoints survived FDR. Effective reachability dimension and directional entropy were nominally higher under hypoglycemia, but neither survived the broader reachability family correction. Reachability log-volume was not significant. These data do not support a confirmed reachability collapse.

Endpoint	HYPO - EU	p	FDR q
m median	−0.743	0.0002	0.00090
m_a median	−1.096	0.0002	0.0018
m_e median	−0.685	0.0004	0.0018
d_n median	1.014	0.0008	0.0031
e_e median	−0.708	0.0176	0.0396
e_m median	1.024	0.0016	0.0048
Reachability dimension	0.251	0.0124	0.134
Reachability log-volume	0.395	0.245	0.537

Table 2: Selected paired endpoint results. Full engine-level tables are in Supplementary Data S2.

4.4. R4: Label-swap null and adjusted-geometry sensitivity

The 10,000-draw paired label-swap null supported lower m_a, m_e, and e_e, and higher d_n and e_m, after FDR across the prespecified 11-endpoint robustness set. The same restricted null also supported higher reachability dimension and directional entropy. The latter two findings remain qualified because they did not survive the broader engine-level reachability FDR family.

Excluding the three adjusted-geometry hypoglycemia sessions preserved the direction of every prespecified 9D and reachability effect. This directional check does not establish restricted-sample significance or resolve the pooled anchor limitation.

4.5. R5: Feature-level sanity check

Raw feature summaries replicated higher theta ($\Delta = 0.130$, $q = 1.3 \times 10^{-5}$) and delta ($\Delta = 0.102$, $q = 8.6 \times 10^{-5}$) and lower exported sample/permutation entropy ($\Delta = -0.028$, $q = 0.043$). Alpha was also higher. The exported sample- and permutation-entropy fields had identical values in this pipeline surface and should not be treated as independent replications.

Dissociation finding

Hypoglycemia increased low-frequency power while the fixed stratified chart showed lower mobility- and entropy-related components; this is a condition-associated dissociation, not evidence that either layer indexes conscious level.

5. Discussion

5.1. Primary finding

The central empirical result is a stable paired shift on a fixed stratified neural-dynamics surface. Lower coarse mobility was decomposed into lower m_a and m_e, while lower e_e alongside higher e_m shows why a single scalar complexity narrative is inadequate. This extends feature-level hypoglycemia sensitivity in the source study (Rubega et al., 2020) without claiming that the chart reconstructs a unique metabolic neural state.

5.2. What the analysis does not show

The analysis does not show a reachability collapse. Its nominal and restricted-null reachability effects point in the opposite direction from the initial collapse prediction, while the broad engine-level correction is not significant. Local capacity proxies cannot be equated with occupancy, behavior, cognition, or consciousness.

The excerpts are not a continuous metabolic trajectory. The data therefore cannot identify a glucose threshold, rate-of-fall effect, recovery hysteresis, autonomic warning signal, impaired-awareness subgroup, or metabolic adequacy gate.

5.3. Implications for metabolic perturbation studies

The result establishes a bounded condition-associated stratified EEG signature, not a metabolic-anchoring mechanism. Its principal theoretical implication is negative: increased low-frequency power and altered complexity cannot, by themselves, be interpreted as stronger functional coordination or reduced neural capacity. A future study may test metabolic availability as a slow constraint on neural dynamics, but it would require continuous glucose, recovery, behavioral performance, and interoceptive measurements. An euglycemia-only feature anchor is also required before interpreting absolute position relative to a metabolic reference condition.

5.4. Limitations

The data provide 34 paired excerpts but no subject-level clinical covariates or continuous metabolic measurements. The cohort anchor pools both conditions, the recording segments are short, and regional exports are unavailable. The results are condition associations under a specific preprocessing and chart contract. They should not be generalized to untreated hypoglycemia, recovery, clinical severity, or conscious experience.

5.5. Conclusion

Hypoglycemia produced a reproducible paired change in a fixed stratified EEG state-space chart and replicated low-frequency and entropy-related feature changes. The data do not support a confirmed reachability collapse or an embodied-anchoring mechanism. This negative boundary is substantive: neural dynamics under metabolic challenge should be assessed with separable coordinate, local-flow, and reachability layers rather than a single rhythm or complexity score.

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Author Contributions

RL: Conceptualization, Methodology, Data Curation, Analysis, Writing, and Editing.

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Data Availability Statement

The source EEG dataset is available through Zenodo record 3465213. The H5-derived analysis wrappers and result tables are in this repository under `articles/EAP_metabolic_anchoring/`.

Competing Interests

The author declares no competing interests.